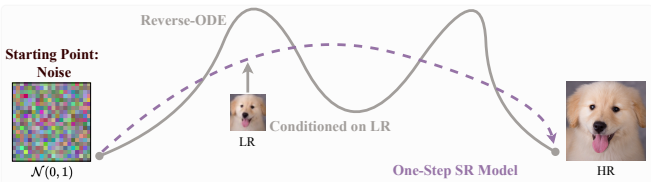


(a) **Vanilla One-Step Diffusion for Real-ISR.** The HR image is directly restored from the LR input in a single step, improving efficiency but sacrificing stochasticity and perceptual quality.



(b) **Our Noise-Started One-Step Diffusion for Real-ISR.** Starting from random Gaussian noise, our method performs one-step HR generation with LR conditioning, achieving efficient inference and strong perceptual quality.

Figure 1. Unlike many diffusion-based one-step methods that directly map LR inputs to HR outputs, our method preserves the noise-started generation paradigm and generates HR images from random noise in a single step under LR conditioning. *This design retains the stochastic generative capacity of diffusion models while enabling fast inference.*